Technische Hochschule
mannheim

Fakultät für
Maschinenbau

Mathematics

Lecture Notes

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Semester WS 26/27

How to Use These Notes

These lecture notes are available both as a complete edition and as a fill-in-the-blank edition. Both editions have the same structure and cover the same mathematical material. In the fill-in-the-blank edition, selected results are replaced by writing lines or grids for working through longer calculations.

The notes accompany the lectures. Derivations, examples, and supplementary comments can be added directly in the spaces provided. Exercises and their discussion form a separate component of the course.

These notes are prepared for courses at Mannheim University of Applied Sciences. Feedback on typographical errors or mathematical ambiguities is always welcome.

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Greek Alphabet

Uppercase	Lowercase	Name	Uppercase	Lowercase	Name
A	α	alpha	N	ν	nu
B	β	beta	Ξ	ξ	xi
Γ	γ	gamma	O	o	omicron
Δ	δ	delta	Π	π	pi
E	ϵ	epsilon	P	ρ	rho
Z	ζ	zeta	Σ	σ	sigma
H	η	eta	T	τ	tau
Θ	θ	theta	Υ	υ	upsilon
I	ι	iota	Φ	ϕ	phi
K	κ	kappa	X	χ	chi
Λ	λ	lambda	Ψ	ψ	psi
M	μ	mu	Ω	ω	omega

Recommended Reading

1. G. Glaeser: *Der mathematische Werkzeugkasten*. Springer Spektrum, 2014.
2. F. Paech: *Analysis – anschaulich und anwendungsorientiert*. Hanser, 2013.
3. K. Meyberg and P. Vachenauer: *Höhere Mathematik 1 – Differential- und Integralrechnung, Vektor- und Matrizenrechnung*. Springer, 2003.
4. L. Papula: *Mathematik für Ingenieure und Naturwissenschaftler, Band 1*. Vieweg, 2009.
5. L. Papula: *Mathematik für Ingenieure und Naturwissenschaftler, Band 2*. Vieweg, 2009.
6. J. Koch and M. Stämpfle: *Mathematik für das Ingenieurstudium*. Hanser, 2010.

The lecture notes and the accompanying exercises, including any provided model solutions, are recommended as supplementary material.

1 Fundamentals